



NORTH SYDNEY BOYS HIGH SCHOOL

2009 PRELIMINARY EXAMINATION

Mathematics Extension 1

Examiner: R. Lowe

General Instructions

- Working time – 2 hours and a half
- Attempt all questions
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new **Question** is to be started on a **new page**.

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
- ☐ Mr Ireland
- ☐ Mr Lam
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Name:

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	8	9	10	11	Total	Total
Mark	10	11	12	12	5	7	13	11	12	12	13	118	100

Question 1

- a) Evaluate $\frac{3.24^2 - 2.1^2}{\sqrt{36 + 2.1}}$ correct to 3 significant figures. 1
- b) Simplify $\frac{2x^2 - 1}{1 - 2x^2}$ 1
- c) Simplify $\frac{3^{n+2} + 3^n}{3^n}$ 2
- d) Given that $\tan \alpha = 2$ where α is an acute angle, find the value of $\tan 2\alpha$ in exact form. 2
- e) Solve $\frac{4}{x-3} \leq 2$ 3
- f) Find correct to 2 decimal places $\log_3 12$ 1

Question 2 Start a new page

- a) Find the acute angle between the lines $4x - y = 0$ and $y = x$ to the nearest degree. 2
- b) Solve $|15 + 4x| \leq 3$ 2
- c) Given that R(x, y) divides the interval joining P(-4, 5) and Q(-1, 9) externally in the ratio 3: 5, find the coordinates of R. 2
- d) Find the period and amplitude for the graph of $y = 3\sin(2\pi x)$. 2
- e) i) Without calculus sketch the graph $y = x^3(2 - x)$ 2
 ii) Hence, or otherwise, solve the inequality $x^3(2 - x) < 0$. 1

Question 3 Start a new page

a) Write $0.\dot{1}\dot{2}$ as a fraction in simplest form 2

b) Simplify fully $\frac{5x+1}{(x+1)(x-1)} + \frac{2}{1-x} - \frac{3}{1+x}$ 2

c) i) Prove that $(\operatorname{cosec}^2 x - 1)\sin^2 x = \cos^2 x$ 2

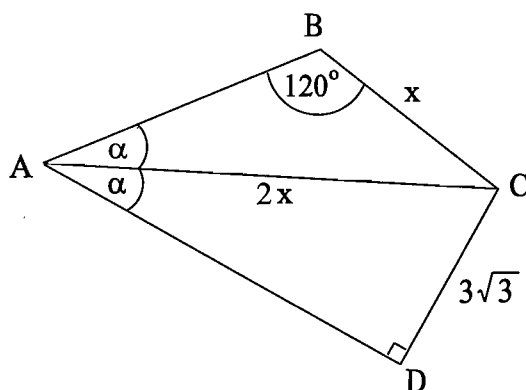
ii) Hence or otherwise solve: 3

$$(\operatorname{cosec}^2 x - 1)\sin^2 x = \frac{3}{4} \quad \text{for } 0 \leq x \leq 360^\circ$$

d)

In the quadrilateral ABCD shown in the diagram, $\angle BAC = \angle CAD = \alpha$.

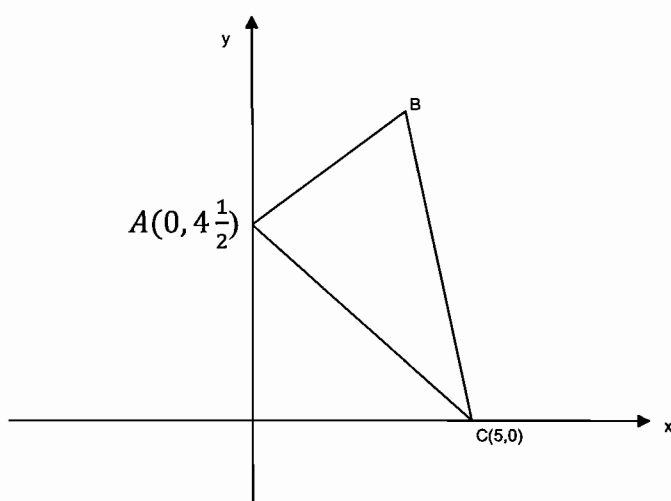
$\angle ABC = 120^\circ$, $\angle ADC = 90^\circ$, $BC = x$, $AC = 2x$, and $CD = 3\sqrt{3}$ cm



i) Use the triangle ABC to find the value of $\sin \alpha$. 2

ii) Hence, find the value of x . 1

Question 4 [Start a new page](#)



The lines AB and CB have equations $x - 2y + 9 = 0$ and $4x - y - 20 = 0$ respectively.

- i) Find the coordinates of the point B . 3
- ii) Show that the equation of the line AC is $9x + 10y - 45 = 0$. 2
- iii) Calculate the distance AC in exact form. 2
- iv) Find the equation of the line perpendicular to BC which passes through A . 2
- v) Calculate the shortest distance between the point B and the line AC . 3

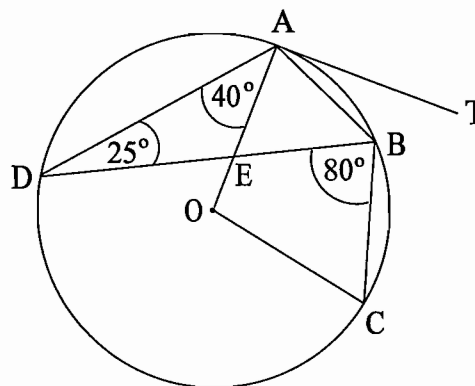
Hence find the area of the triangle ABC .

Question 5 [Start a new page](#)

A, B, C , and D are four points on a circle with centre O , such that

$$\angle ADB = 25^\circ, \angle DBC = 80^\circ \text{ and } \angle OAD = 40^\circ.$$

AT is a tangent at A to the circle.



- i) Show that the triangle AEB is isosceles. 3
- ii) Show that quadrilateral $OEBC$ is cyclic. 2

Question 6 Start a new page

- a) The first and last terms of an arithmetic series are 5 and 165 respectively, and the sum of these terms is 1785. Find the number of terms in the series. 2
- b) i) Write down a single expression for the sum of the first n terms of the geometric series $1 + x^2 + x^4 + x^6 + \dots$ 2
- ii) When does the sum to infinity exist (in terms of x^2) ? 1
- iii) What is the sum to infinity when $x = \frac{1}{3}$? 2

Question 7 Start a new page

- a) Differentiate with respect to x
- i) $y = 4x^8 - \frac{1}{\sqrt{x}}$ 2
- ii) $y = (3x^2 + 7)^8$ 2
- iii) $y = \frac{x^3}{x^2+1}$ 2
- b) A particle moves in a straight line so that its position, x m, from the origin at time, t seconds, is given by $x = t^3 + 3t^2 - 9t$
- Find
- i) its initial displacement, velocity and acceleration. 3
- ii) when the particle is stationary and its position at that time. 2
- iii) the distance travelled in the first 2 seconds. 2

Question 8 Start a new page

- a) For the curve $y = x^3 + 3x^2 - 24x + 2$
- i) Find the stationary points and determine their nature 3
- ii) Find any points of inflexion 2
- iii) Sketch the curve 2
- b) A farmer has 30m of wire fencing. He wishes to make a rectangular enclosure using the side of the barn as one wall. What is the largest area that the fenced enclosure can have ? Justify your answer by using calculus. 4

Question 9 **Start a new page**

- a) If $\sin \phi = \frac{4}{5}$ and $\sin \alpha = \frac{5}{13}$, find $\cos (\phi - \alpha)$ if α and ϕ are acute 3
- b) Find the exact value of $\sin 105^\circ$. Express your answer with a rational denominator. 3
- c) Solve $\cos \left(2x - \frac{\pi}{6} \right) = \frac{\sqrt{3}}{2}$ where $0 \leq x \leq 2\pi$ 3
- d) A ship is sailing on a bearing of 205° from port A to port B. If port B is 70 nautical miles west of port A, find how far the ship has sailed. 3

Question 10 **Start a new page**

- a) Consider the function $y=f(x)$. If the point (4,8) is a maximum turning point on the function what is the value of $f'(4)$? 1
- b) On the same curve $y=f(x)$ a tangent is drawn from the point (3,6) and this tangent has a y intercept of 10. What is the value of $f'(3)$? 2
- c) On separate axes sketch the following graphs, showing all important features, and state their domain and range
- i) $y = 4\sin 2x$ $0^\circ \leq x \leq 360^\circ$ 3
- ii) $y = \sqrt{4 - x^2}$ 3
- iii) $y = 4^{-x}$ 3

Question 11 **Start a new page**

- a) Points A, B and C lie on horizontal ground so that $\angle ACB$ is 90° and A and B are 200m apart. Point D is h metres vertically above C. The angles of elevation of D from A and B are 20° and 23° respectively. Draw a half page diagram to show this information and find the value of h correct to 2 significant figures. 4

b) Solve $\log_2 x - \log_2(x-2) = \frac{2}{3} \log_2 27$ 3

- c) The radius of curvature R of a function $y = f(x)$ is a measure of how sharply a curve is bending.

It is defined as
$$R = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}.$$

Use this formula to find the exact radius of curvature of the curve $y = \sqrt{x}$ at $x = 1$. 3

- d) Find the domain and range of the function $f(x) = \sqrt{3 - \sqrt{x}}$ 3

Suggested Solutions

Question 1 (Weiss)

(a) (1 mark)

$$\frac{3.24^2 - 2.1^2}{\sqrt{36 + 2.1}} = 0.986$$

(b) (1 mark)

$$\frac{2x^2 - 1}{1 - 2x^2} = -1$$

(c) (2 marks)

$$\frac{3^{n+2} + 3^n}{3^n} = \frac{\cancel{3^2}(3^2 + 1)}{\cancel{3^0}} = 9 + 1 = 10$$

(d) (2 marks)

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

Substitute $\tan \alpha = 2$ results in

$$\tan 2\alpha = \frac{2 \times 2}{1 - 2^2} = -\frac{4}{3}$$

(e) (3 marks)

$$\frac{4}{x-3} \leq \frac{2}{x(x-3)^2}$$

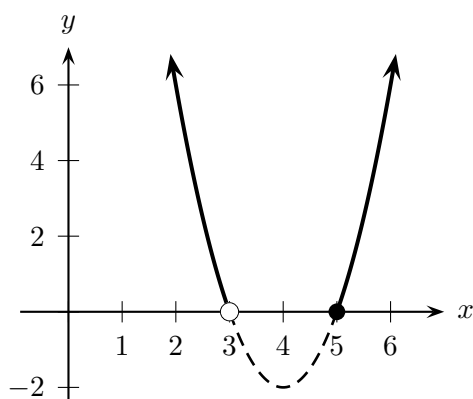
$$4(x-3) \leq 2(x-3)^2$$

$$2(x-3)^2 - 4(x-3) \geq 0$$

$$2(x-3)(x-3-2) \geq 0$$

$$2(x-3)(x-5) \geq 0$$

$$x < 3 \text{ or } x \geq 5$$



(f) (1 mark)

$$\log_3 12 = \frac{\log_{10} 12}{\log_{10} 3} = 2.26$$

Question 2 (Lowe)

(a) (2 marks)

$$4x - y = 0 \Rightarrow m_1 = 4$$

$$y = x \Rightarrow m_2 = 1$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{4 - 1}{1 + 4} \right| = \frac{3}{5}$$

$$\therefore \theta = \tan^{-1} \frac{3}{5} = 30^\circ 58'$$

(b) (2 marks)

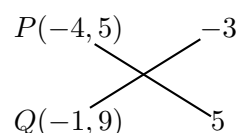
$$|15 + 4x| \leq 3$$

$$-3 \leq 15 + 4x \leq 3$$

$$\div 4 \quad -18 \leq 4x \leq -12$$

$$-\frac{9}{2} \leq x \leq -3$$

(c) (2 marks)



As R divides PQ externally,

$$R(x, y) = \left(\frac{5(-4) + -3(-1)}{-3 + 5}, \frac{5(5) + -3(9)}{-3 + 5} \right)$$

$$= \left(-\frac{17}{2}, -\frac{2}{2} \right) = \left(-8\frac{1}{2}, -1 \right)$$

(d) (2 marks)

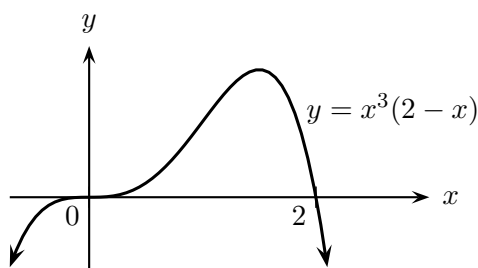
$$y = 3 \sin(2\pi x)$$

- amplitude $a = 3$.
- period:

$$f = \frac{2\pi}{T}$$

$$f = 2\pi \Rightarrow T = \frac{2\pi}{2\pi} = 1$$

(e) i. (2 marks)



ii. (1 mark)

$x^3(2 - x)$ is below the x axis when
 $x < 0$ or $x > 2$.

ii. (3 marks)

$$(\operatorname{cosec}^2 x - 1) \sin^2 x = \frac{3}{4}$$

$$\cos^2 x = \frac{3}{4}$$

$$\cos x = \pm \frac{\sqrt{3}}{2}$$

$$x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

(d) i. (2 marks)

Applying the sine rule in $\triangle ABC$,

$$\frac{\sin \alpha}{x} = \frac{\sin 120^\circ}{2x}$$

$$\therefore \sin \alpha = \frac{\sin 120^\circ}{2} = \frac{\sqrt{3}}{4}$$

Question 3 (Rezcallah)

(a) (2 marks)

$$\begin{aligned} x &= 0.122 \dots \\ 10x &= 1.222 \dots \\ 100x &= 12.222 \dots \\ \hline 90x &= 11 \\ \therefore x &= \frac{11}{90} \end{aligned}$$

(b) (2 marks)

$$\begin{aligned} & \frac{5x+1}{(x+1)(x-1)} + \frac{2}{1-x} - \frac{3}{1+x} \\ &= \frac{5x+1}{(x+1)(x-1)} - \frac{2}{x-1} - \frac{3}{x+1} \\ &= \frac{(5x+1) - 2(x+1) - 3(x-1)}{(x+1)(x-1)} \\ &= \frac{5x+1-2x-2-3x+3}{(x+1)(x-1)} \\ &= \frac{2}{(x+1)(x-1)} \text{ or } -\frac{2}{(1-x)(x+1)} \end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned} & (\operatorname{cosec}^2 x - 1) \sin^2 x \\ &= \left(\frac{1}{\sin^2 x} - 1 \right) \sin^2 x \\ &= 1 - \sin^2 x \\ &= \cos^2 x \end{aligned}$$

ii. (1 mark)

$$\text{In } \triangle ACD, \sin \alpha = \frac{3\sqrt{3}}{2x}$$

$$\text{But } \sin \alpha = \frac{\sqrt{3}}{4},$$

$$\begin{aligned} \therefore \frac{\sqrt{3}}{4} &= \frac{3\sqrt{3}}{2x} \\ 2x &= 12 \\ \therefore x &= 6 \end{aligned}$$

Question 4 (Lam)

(a) (3 marks)

$$\begin{cases} x - 2y + 9 = 0 & (1) \\ 4x - y - 20 = 0 & (2) \\ 8x - 2y - 40 = 0 & (2') \end{cases}$$

Subtract (2') from (1),

$$-7x + 49 = 0$$

$$7x = 49$$

$$x = 7 \rightarrow (1)$$

$$7 - 2y + 9 = 0$$

$$2y = 16$$

$$y = 8$$

$$\therefore B(7, 8)$$

(b) (2 marks)

$$\begin{aligned}
 m_{AC} &= \frac{\frac{9}{2} - 0}{0 - 5} = -\frac{9}{10} \\
 \frac{y - \frac{9}{2}}{x - 0} &= -\frac{9}{10} \\
 \frac{y}{\times 10} &= -\frac{9}{\times 10} + \frac{9}{\times 10} \\
 10y &= -9x + 45 \\
 9x + 10y - 45 &= 0
 \end{aligned}$$

(c) (2 marks)

$$\begin{aligned}
 d_{AC} &= \sqrt{5^2 + \left(\frac{9}{2}\right)^2} \\
 &= \sqrt{\frac{181}{4}} \\
 &= \frac{\sqrt{181}}{2}
 \end{aligned}$$

(d) (2 marks)

$$\begin{aligned}
 m_{BC} &= 4 \\
 y &= -\frac{1}{4}x + \frac{9}{2}
 \end{aligned}$$

(e) (3 marks)

Perpendicular distance from $(7, 8)$ to $9x + 10y - 45 = 0$ is

$$\begin{aligned}
 d_{\perp} &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\
 &= \frac{|9(7) + 10(8) - 45|}{\sqrt{9^2 + 10^2}} \\
 &= \frac{|63 + 80 - 45|}{\sqrt{181}} \\
 &= \frac{98}{\sqrt{181}} \\
 A &= \frac{1}{2}bh \\
 &= \frac{1}{2} \times \frac{\sqrt{181}}{2} \times \frac{98}{\sqrt{181}} \\
 &= \frac{98}{4} = \frac{49}{2} \\
 &= 24.5 \text{ units}^2
 \end{aligned}$$

Question 5 (Trenwith)

(a) (3 marks)

- $\angle TAB = 25^\circ$ (angle between a chord and a tangent is equal to the angle in the alternate segment).
 - $\angle OAT = 90^\circ$ (angle between a tangent and a radius).
 - $\therefore \angle OAB = 65^\circ$ (by subtraction).
 - $\angle AEB = 40^\circ + 25^\circ = 65^\circ$.
(exterior angle of $\triangle ADE$ equals the sum of the opposite interior angles).
- $\therefore \triangle AEB$ isosceles as it has 2 equal angles.

(b) (2 marks)

- $\angle ABE = 50^\circ$ (angle sum of $\triangle ABE$)
- $\therefore \angle ABC = 50^\circ + 80^\circ = 130^\circ$.

$$\therefore \text{reflex } \angle AOC = 260^\circ$$

(Angle at the centre is twice the angle at the circumference subtended by the same arc AC .)

- Hence $\angle AOC = 100^\circ$ (angle of revolution at the centre O).
- $\therefore OEBC$ is a cyclic quadrilateral as its opposite angles $\angle COE$ & $\angle EBC$ are supplementary.

Question 6 (Barrett)

(a) (2 marks)

$$\begin{aligned}
 a &= 5 & \ell &= a + (n - 1)d = 165 \\
 S_n &= 1785 \\
 1785 &= \frac{n}{2}(a + \ell) \\
 3570 &= n(5 + 165) \\
 n &= \frac{3570}{170} = 21
 \end{aligned}$$

(b) i. (2 marks)

$$\begin{aligned}
 a &= 1 & r &= x^2 \\
 S_n &= \frac{a(r^n - 1)}{r - 1} \\
 &= \frac{(x^2)^n - 1}{x^2 - 1} \\
 &= \frac{x^{2n} - 1}{x^2 - 1}
 \end{aligned}$$

ii. (1 mark)

$$\begin{aligned}
 |r| &< 1 \\
 |x^2| &< 1 \\
 \therefore 0 &\leq x^2 < 1 \\
 \therefore -1 &< x < 1
 \end{aligned}$$

iii. (2 marks)

$$\begin{aligned}
 S &= \frac{a}{1-r} = \frac{a}{1-x^2} \\
 &= \frac{1}{1-\left(\frac{1}{3}\right)^2} = \frac{1}{1-\frac{1}{9}} = \frac{9}{8}
 \end{aligned}$$

Question 7 (Barrett)

(a) i. (2 marks)

$$\begin{aligned}
 y &= 4x^8 - x^{-\frac{1}{2}} \\
 \frac{dy}{dx} &= 32x^7 + \frac{1}{2}x^{-\frac{3}{2}}
 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned}
 y &= (3x^2 + 7)^8 \\
 \frac{dy}{dx} &= 8 \times 6x (3x^2 + 7)^7 \\
 &= 48x (3x^2 + 7)^7
 \end{aligned}$$

iii. (2 marks)

$$\begin{aligned}
 y &= \frac{x^3}{x^2 + 1} \\
 u &= x^3 \quad v = x^2 + 1 \\
 u' &= 3x^2 \quad v' = 2x \\
 \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\
 &= \frac{(3x^2)(x^2 + 1) - (2x)(x^3)}{(x^2 + 1)^2} \\
 &= \frac{3x^4 + 3x^2 - 2x^4}{(x^2 + 1)^2} \\
 &= \frac{x^4 + 3x^2}{(x^2 + 1)^2}
 \end{aligned}$$

(b) i. (3 marks)

$$\begin{aligned}
 x(t) &= t^3 + 3t^2 - 9t \\
 v(t) &= 3t^2 + 6t - 9 \\
 a(t) &= 6t + 6 \\
 x(0) &= 0 \quad v(0) = -9 \quad a(0) = 6
 \end{aligned}$$

ii. (2 marks) Particle is stationary when $v(t) = 0$.

$$\begin{aligned}
 3t^2 + 6t - 9 &= 0 \\
 t^2 + 2t - 3 &= 0 \\
 (t + 3)(t - 1) &= 0
 \end{aligned}$$

As $t > 0$,

$$\begin{aligned}
 t &= 1 \\
 x(1) &= 1 + 3 - 9 = -5
 \end{aligned}$$

The particle is 5 units to the left of the origin when it is stationary.

iii. (2 marks)

$$\begin{aligned}
 x(0) &= 0 \\
 x(1) &= 1 + 3 - 9 = -5 \\
 x(2) &= 8 + 12 - 18 = 2
 \end{aligned}$$

The particle travels -5 units between $t = 0$ and $t = 1$, then from $t = 1$ to $t = 2$ travels from -5 to 2 (7 units).
Total distance travelled: $5 + 7 = 12$ units.

Question 8 (Trenwith)

(a) i. (3 marks)

$$\begin{aligned}
 y &= x^3 + 3x^2 - 24x + 2 \\
 y' &= 3x^2 + 6x - 24 \\
 y'' &= 6x + 6
 \end{aligned}$$

Stationary pts occur when $y' = 0$,

$$\begin{aligned}
 3x^2 + 6x - 24 &= 0 \\
 x^2 + 2x - 8 &= 0 \\
 (x + 4)(x - 2) &= 0 \\
 \therefore x &= -4, 2
 \end{aligned}$$

$$\begin{aligned}
 y'' = 6x + 6 \Big|_{x=-4} & \quad \quad \quad y'' = 6x + 6 \Big|_{x=2} \\
 = 6(-4) + 6 < 0 & \quad \quad \quad = 6(2) + 6 > 0
 \end{aligned}$$

Hence $x = -4$ is a local maximum and $x = 2$ is a local minimum.

x	-4	2
y'	+ 0 -	0 +
y	82	-26

ii. (2 marks)

Points of inflexion occur when $y'' = 0$
with a change in concavity:

$$y'' = 6x + 6 = 0$$

$$6x + 6 = 0$$

$$6x = -6$$

$$x = -1$$

$$y'' = 6x + 6 \Big|_{x=-2} \quad \Bigg| \quad y'' = 6x + 6 \Big|_{x=0}$$

$$< 0 \quad \quad \quad > 0$$

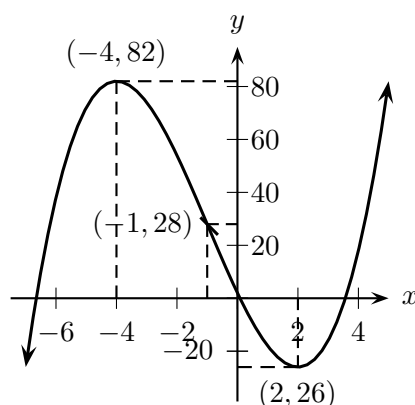
Concavity changes $x = -1$.

$$y = x^3 + 3x^2 - 24x + 2 \Big|_{x=-1}$$

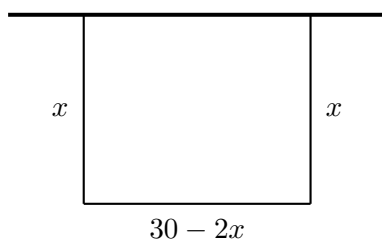
$$= -1 + 3 + 24 + 2 = 28$$

$(-1, 28)$ is a point of inflexion.

iii. (2 marks)



(b) (4 marks)



$$A = x(30 - 2x)$$

$$= 30x - 2x^2$$

$$\frac{dA}{dx} = 30 - 4x$$

$$\frac{d^2A}{dx^2} = -4$$

Maximum area occurs when $\frac{dA}{dx} = 0$

$$30 - 4x = 0$$

$$\frac{4x}{\div 4} = \frac{30}{\div 4}$$

$$x = \frac{15}{2}$$

$\therefore$ maximum area is

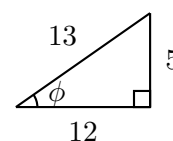
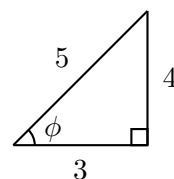
$$A = x(30 - 2x) \Big|_{x=\frac{15}{2}}$$

$$= \left(\frac{15}{2}\right)(30 - 15)$$

$$= \frac{225}{2} \text{ m}^2$$

Question 9 (Lowe)

(a) (3 marks)



$$\sin \phi = \frac{4}{5} \quad \sin \alpha = \frac{5}{13}$$

$$\Rightarrow \cos \phi = \frac{3}{5} \quad \cos \alpha = \frac{12}{13}$$

$$\cos(\phi - \alpha) = \cos \phi \cos \alpha + \sin \phi \sin \alpha$$

$$= \left(\frac{3}{5} \times \frac{12}{13}\right) + \left(\frac{4}{5} \times \frac{5}{13}\right)$$

$$= \frac{36 + 20}{65} = \frac{56}{65}$$

(b) (3 marks)

$$\sin 105^\circ = \sin(60^\circ + 45^\circ)$$

$$= \sin 60^\circ \cos 45^\circ + \sin 45^\circ \cos 60^\circ$$

$$= \left(\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} \times \frac{1}{2}\right)$$

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

(c) (3 marks)

$$\cos\left(2x - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$2x - \frac{\pi}{6} = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

As $x \in [0, 2\pi]$, then $2x \in [0, 4\pi]$.

$$2x - \frac{\pi}{6} = \frac{\pi}{6}, \left(2\pi - \frac{\pi}{6}\right)$$

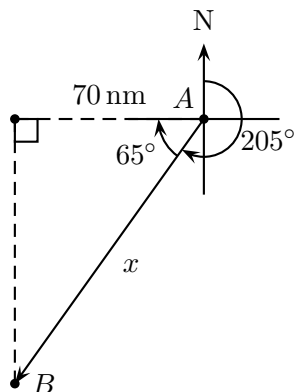
$$\left(2\pi + \frac{\pi}{6}\right), \left(4\pi - \frac{\pi}{6}\right)$$

$$2x = \frac{\pi}{3}, 2\pi, \frac{14\pi}{6}, 4\pi$$

$$x = 0, \frac{\pi}{6}, \pi, \frac{7\pi}{6}, 2\pi$$

 $x = 0$ is inserted as $x = 2\pi$ is also a solution and $x \in [0, 2\pi]$.

(d) (3 marks)



$$\cos 65^\circ = \frac{70}{x}$$

$$x = \frac{70}{\cos 65^\circ} = 165.6 \text{ nm}$$

Question 10 (Weiss)

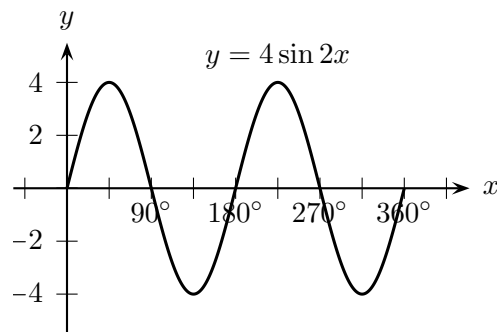
(a) (1 mark)

$$f'(4) = 0$$

(b) (2 marks)

$$f'(3) = \frac{6 - 10}{3 - 0} = -\frac{4}{3}$$

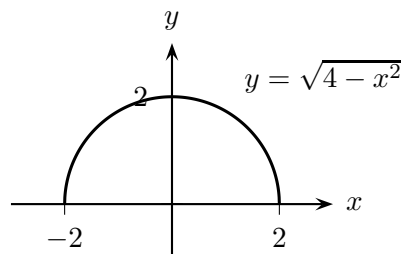
(c) i. (3 marks)



$$D = \{x : 0^\circ \leq x \leq 360^\circ\}$$

$$R = \{y : -4 \leq y \leq 4\}$$

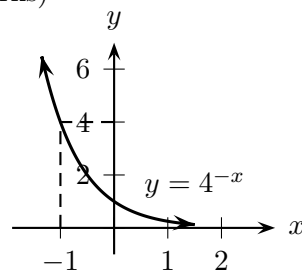
ii. (3 marks)



$$D = \{x : -2 \leq x \leq 2\}$$

$$R = \{y : 0 \leq y \leq 2\}$$

iii. (3 marks)



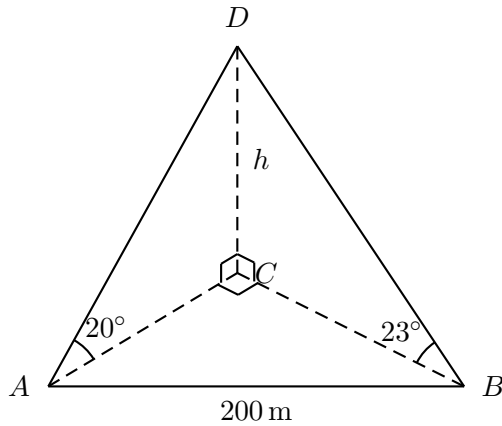
$$D = \{x : x \in \mathbb{R}\}$$

$$R = \{y : y > 0\}$$

Question 11 (Ireland)

(c) (3 marks)

(a) (4 marks)



$$\tan 20^\circ = \frac{h}{AC} \quad \left| \quad \tan 23^\circ = \frac{h}{BC} \right.$$

$$AC = \frac{h}{\tan 20^\circ} \quad \left| \quad BC = \frac{h}{\tan 23^\circ} \right.$$

Using Pythagoras' Theorem in $\triangle ABC$,

$$AC^2 + BC^2 = 200^2$$

$$\frac{h^2}{\tan^2 20^\circ} + \frac{h^2}{\tan^2 23^\circ} = 200^2$$

$$\frac{h^2 (\tan^2 20^\circ + \tan^2 23^\circ)}{\tan^2 20^\circ \tan^2 23^\circ} = \frac{200^2}{\tan^2 20^\circ \tan^2 23^\circ}$$

$$h^2 (\tan^2 20^\circ + \tan^2 23^\circ) = 200^2 (\tan^2 20^\circ \tan^2 23^\circ)$$

$$h^2 = \frac{200^2 (\tan^2 20^\circ \tan^2 23^\circ)}{\tan^2 20^\circ + \tan^2 23^\circ}$$

$$h = \frac{200 (\tan 20^\circ \tan 23^\circ)}{\sqrt{\tan^2 20^\circ + \tan^2 23^\circ}}$$

$$= 55.26$$

$$= 55 \text{ m (2 s.f.)}$$

(b) (3 marks)

$$\log_2 x - \log_2(x-2) = \frac{2}{3} \log_2 27$$

$$\log_2 \left(\frac{x}{x-2} \right) = \log_2 27^{\frac{2}{3}}$$

$$\left(\frac{x}{x-2} \right) = 9$$

$$x = 9x - 18$$

$$\frac{8x}{\div 8} = \frac{18}{\div 8}$$

$$x = \frac{9}{4}$$

$$y = x^{\frac{1}{2}} \quad x = 1$$

$$y' = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

$$y'' = -\frac{1}{4} x^{-\frac{3}{2}}$$

$$R = \frac{\left(1 + \left(\frac{dy}{dx} \right)^2 \right)^{\frac{3}{2}}}{\left| \frac{d^2y}{dx^2} \right|}$$

$$= \frac{\left(1 + \left(\frac{1}{2\sqrt{x}} \right)^2 \right)^{\frac{3}{2}}}{\left| -\frac{1}{4} x^{-\frac{3}{2}} \right|}$$

$$= \frac{\left(1 + \frac{1}{4x} \right)^{\frac{3}{2}}}{\left| -\frac{1}{4} x^{-\frac{3}{2}} \right|} \Bigg|_{x=1}$$

$$= \frac{\left(\frac{5}{4} \right)^{\frac{3}{2}}}{\frac{1}{4}} = \frac{5\sqrt{5}}{8}$$

$$= \frac{5\sqrt{5}}{2}$$

(d) (3 marks)

$$f(x) = \sqrt{3 - \sqrt{x}}$$

As the expression within the radical is ≥ 0

$$3 - \sqrt{x} \geq 0$$

$$3 \geq \sqrt{x}$$

$$0 \leq x \leq 9$$

$$\therefore D = \{x : x \in [0, 9]\}$$

$$f(0) = \sqrt{3} \quad f(9) = 0$$

$$\therefore R = \{y : y \in [0, \sqrt{3}]\}$$